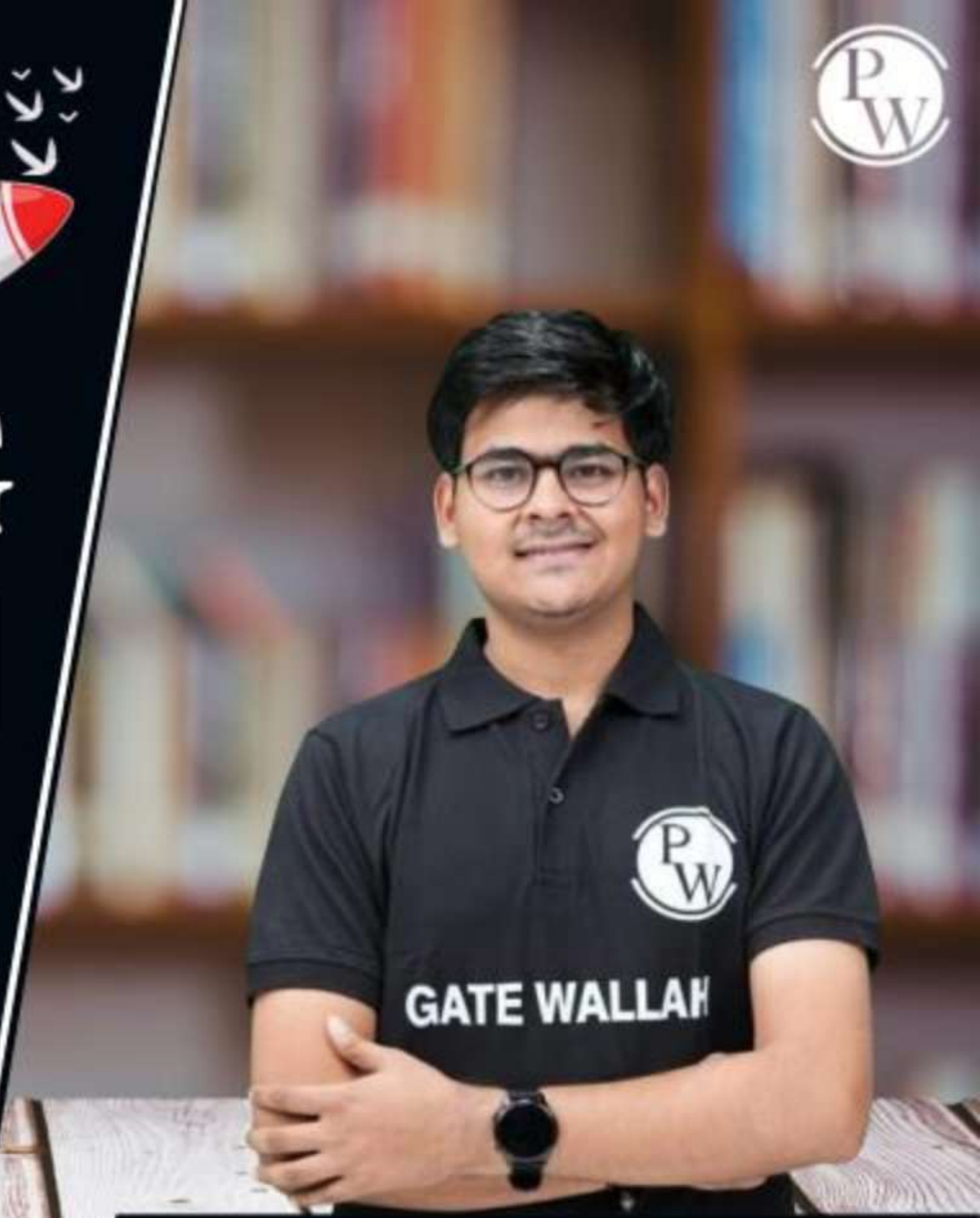
CS & IT ENGINEERING

Digital Logic

Combinational Circuits

Lecture No. 10



By- Aditya sir

Recap of Previous Lecture

Topic

HA
FA
HS
FS →

Topics to be Covered

Topic

Combinational Circuits

- H.S
- F.S
- Serial Adder

Imp Practice Questions.

Topic : Full Adder

Summary

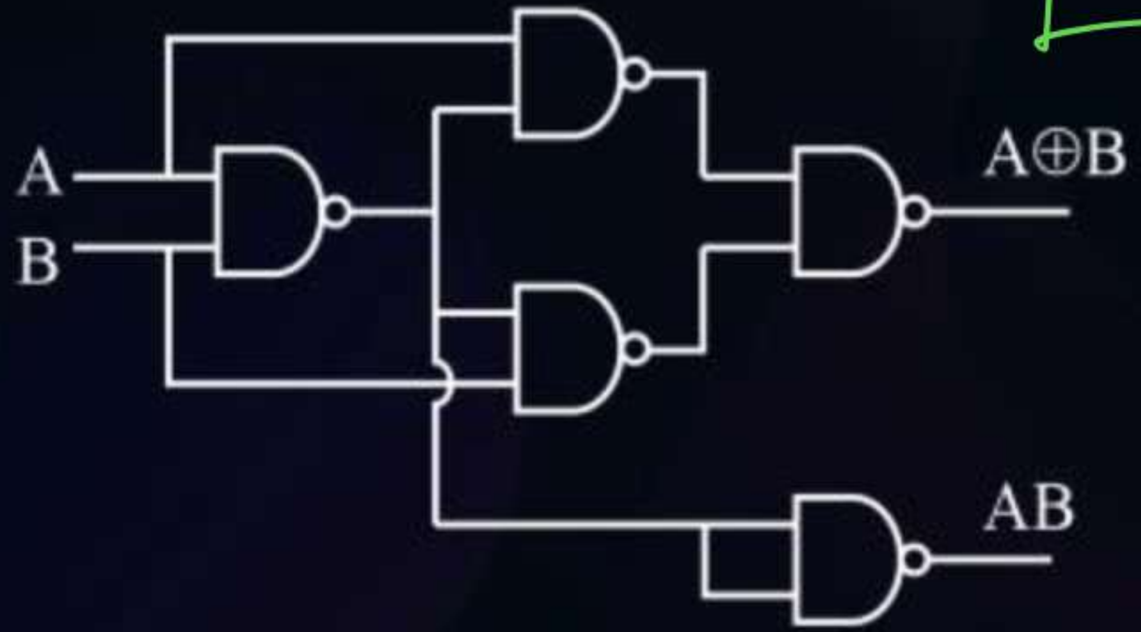
NAND/NOR = 9

Half adder

Sum = $A \oplus B$

Carry = AB

NAND /NOR = 5



Full adder

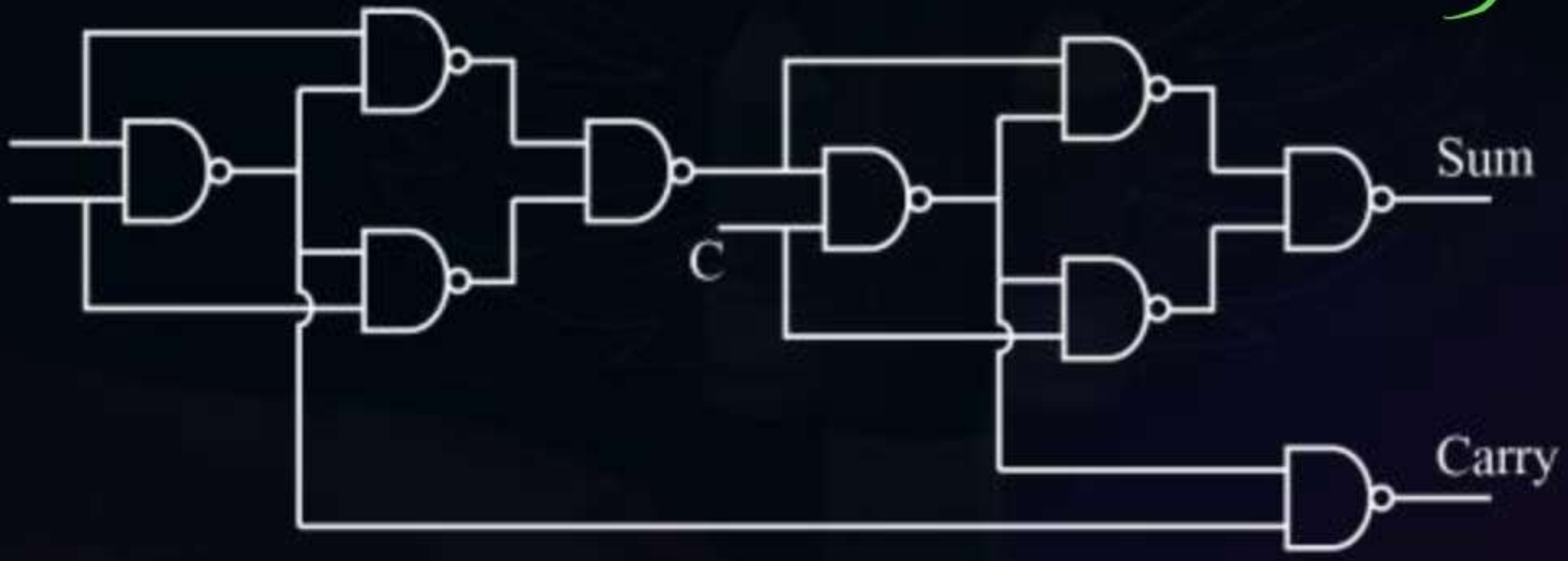
Sum = $\sum m\{1,2,4,7\} = A \oplus B \oplus C$

Carry = $\sum m\ (3,5,6,7)$

$= (A \oplus B)\ C + AB$ → Semi-minimized

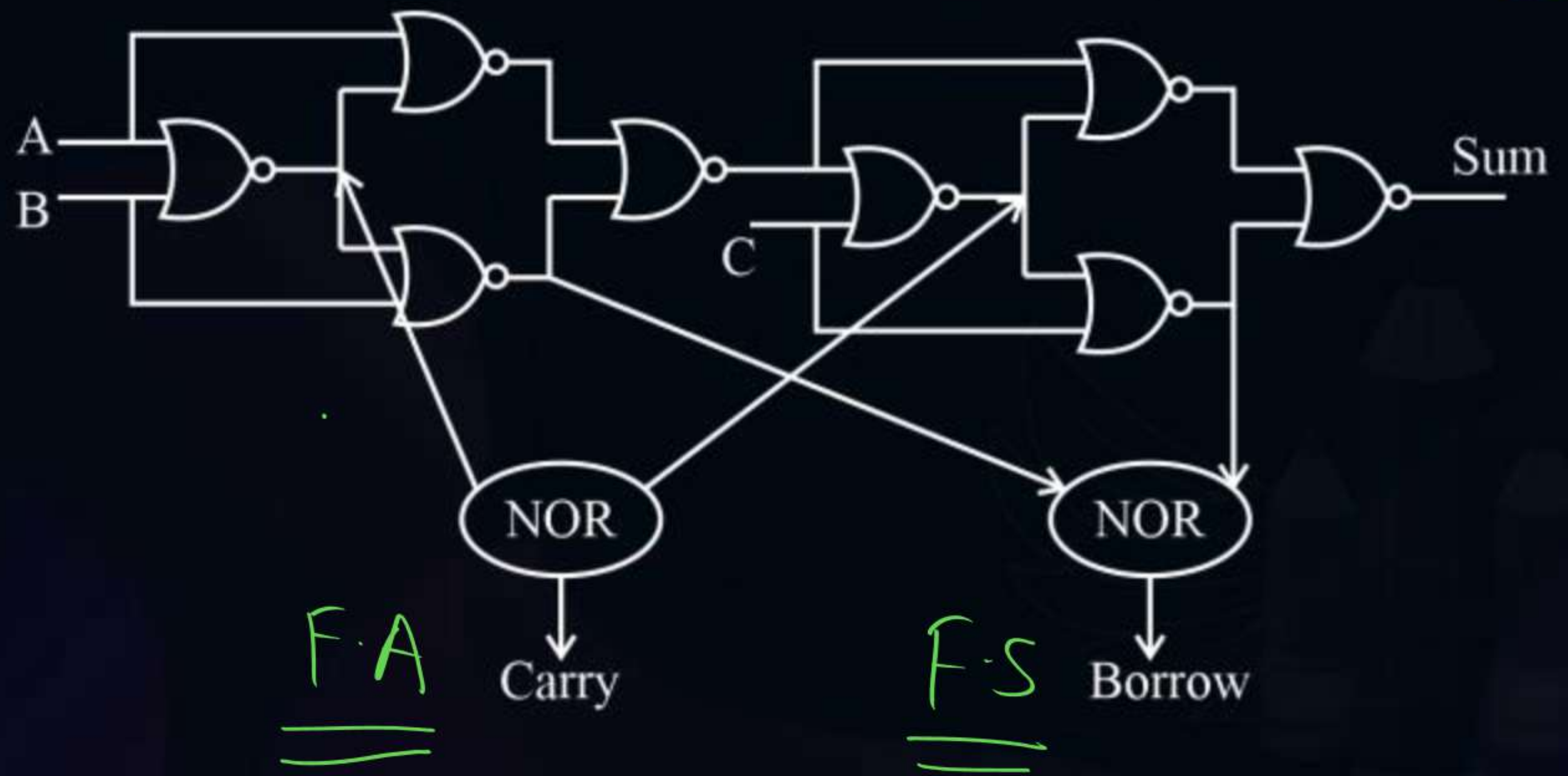
$= AB + BC + AC$

FA = (2HA + OR)



Topic : Full Adder

9 NAND/NOR for FA FS



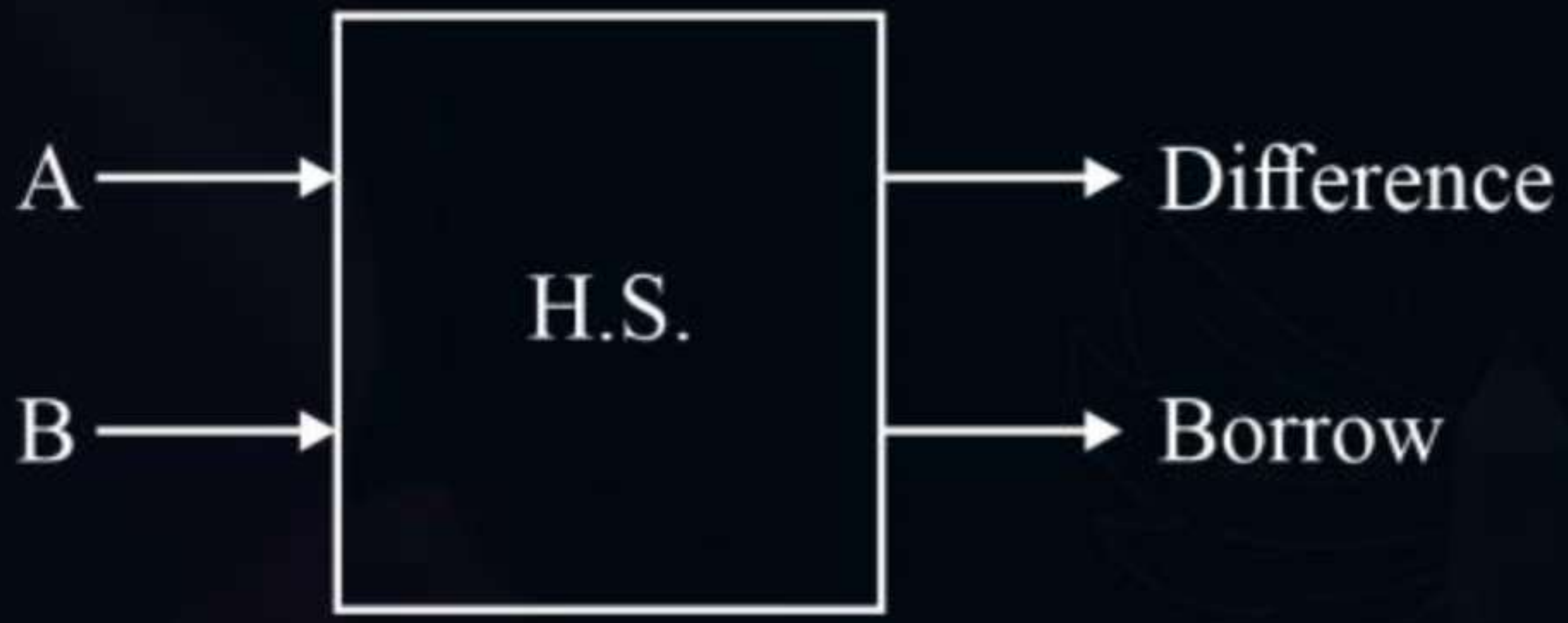
Topic : Half Subtractor

Two bit subtractor are known as half subtractor.

Step-1

2 i/p

2 o/p



Topic : Half Subtractor

Step-1

A	B	Diff.	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$\begin{array}{r} A \\ -\ B \\ \hline \\ \hline \end{array}$$

Step-2

$$\begin{array}{r} 0 \\ -\ 1 \\ \hline 1\ 1 \\ \hline \end{array}$$

Topic : Half Subtractor

Step-3

Diff. $A \oplus B$

Borrow: $\overline{A}B$

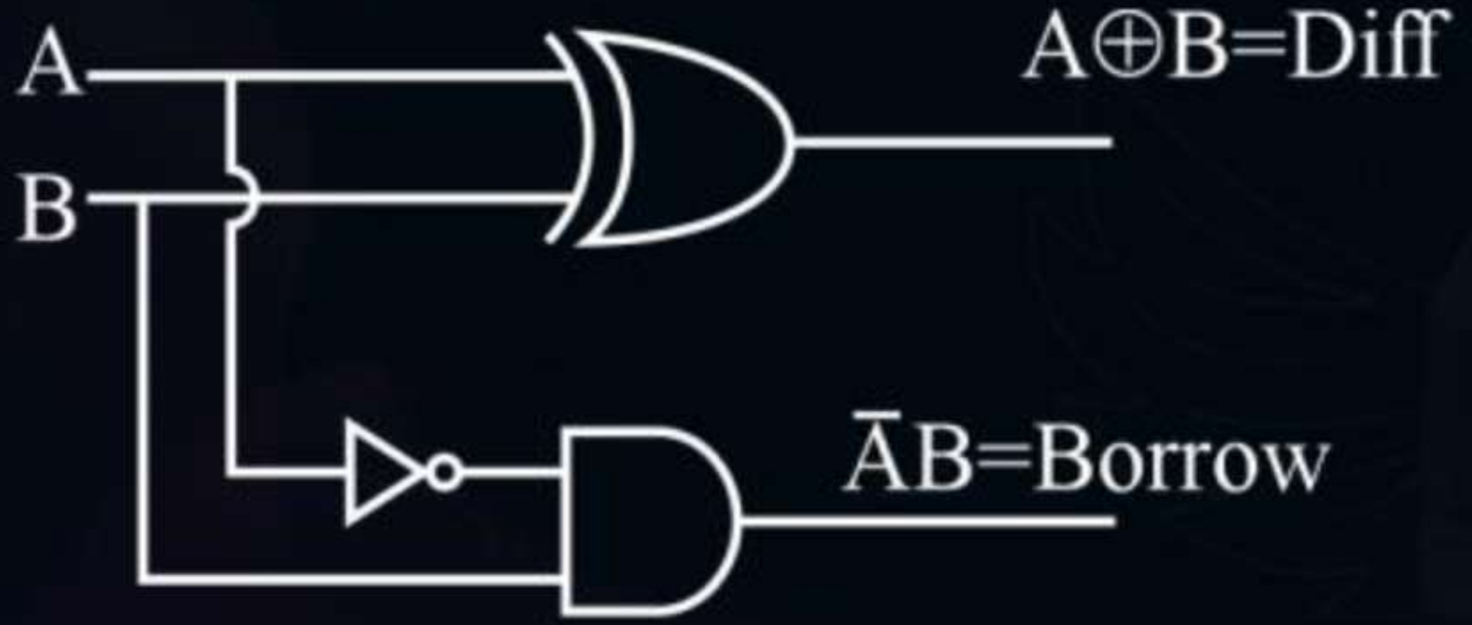
Shortcut FA/HA $A \Rightarrow \overline{A}$ $\Rightarrow$ FS/HS

Topic : Half Subtractor

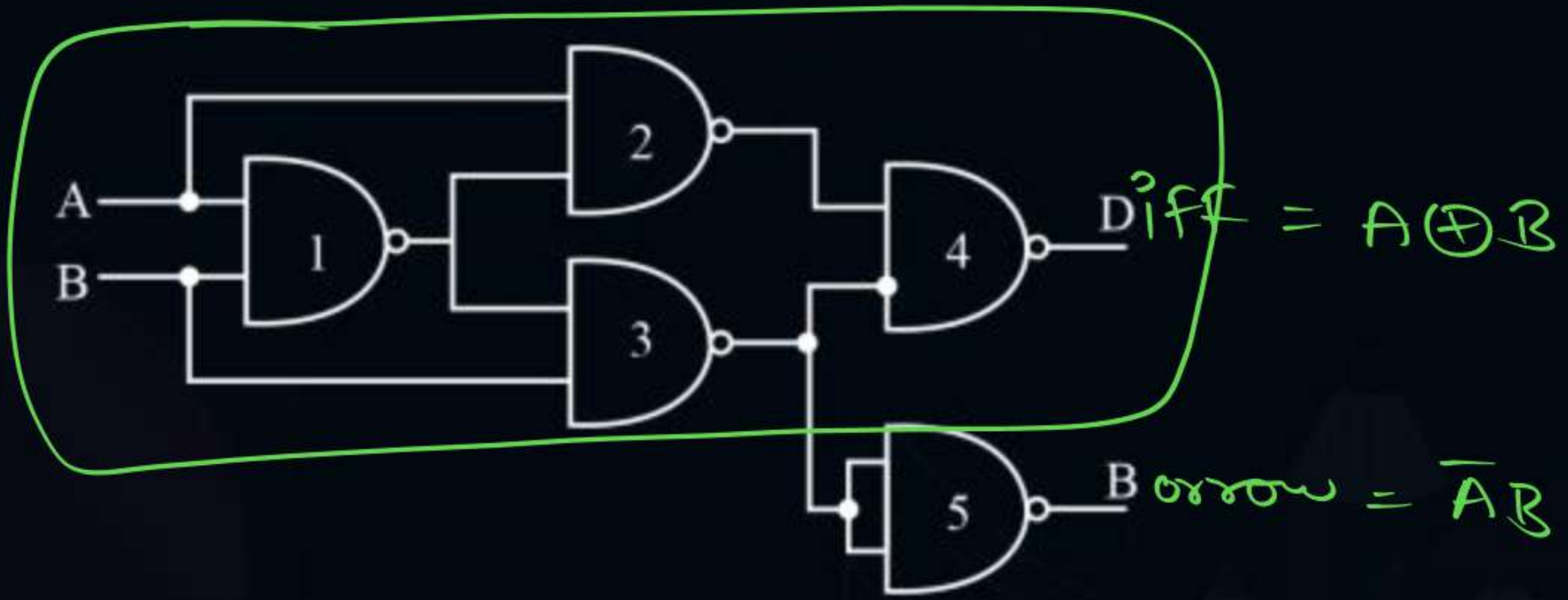
Step-4

No minimization Reqd

Step-5



Topic : Half Subtractor



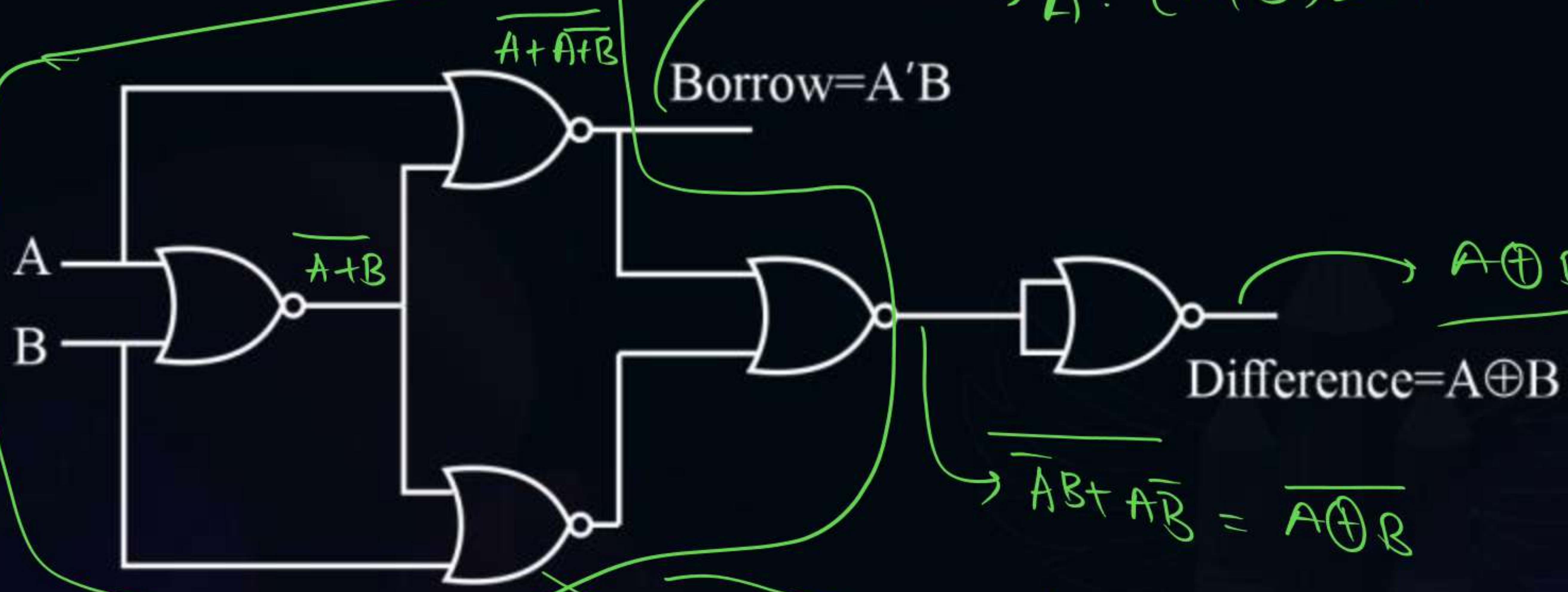
5 NAND

Half Subtractor using NAND gate

Topic : Half Subtractor

$\text{5 NOR} \rightarrow \text{H.S}$

H.S



$\bar{A}\cdot(A+B) = \bar{A}B$

$A \oplus B$

$\overline{\bar{A}B + A\bar{B}} = \overline{A \oplus B}$

$\overline{B+\overline{A+B}} = \bar{B}(A+B) = A\bar{B}$

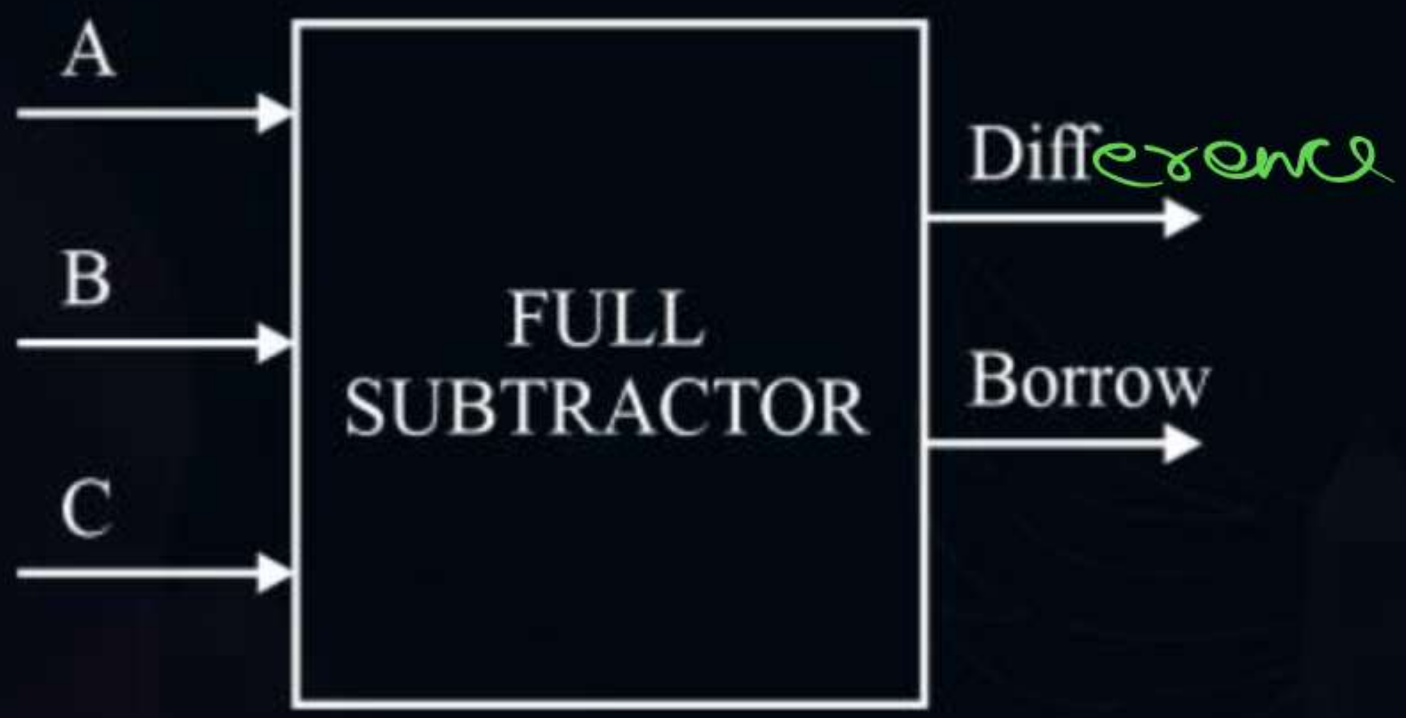
$A \odot B$

Topic : Full Subtractor

Step1 :- i/p & o/p

F.S

↳ 3 bit i/p



3 i/p
2 o/p

Topic : Full Subtractor

Step 2:- TT

E.S

$(A-B)$
↓
Result $- C$

$(A-B)$

Borrow = 1

INPUT			OUTPUT	
A	B	C	Diff	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

0 1 0

A	B	C
0	0	0

$$\begin{array}{r} 0 \\ -\ 0 \\ \hline 0\ 0 \end{array}$$

$$\begin{array}{r} 0 \\ -\ 0 \\ \hline 0\ 0 \end{array} \rightarrow$$

2

A	B	C	Diff	Borrow
0	1	0	1	1

$$\begin{array}{r} 0 \\ -\ 1 \\ \hline 1\ 1 \end{array} \rightarrow$$

$$\begin{array}{r} 1 \\ -\ 0 \\ \hline 1 \end{array} \rightarrow \text{diff}$$

3

A	B	C	Diff	Borr
0	1	1	0	1

$$\begin{array}{r} 0 \\ -\ 1 \\ \hline 1\,1 \end{array}$$

$$\begin{array}{r} 1 \\ -\ 1 \\ \hline 0 \end{array}$$

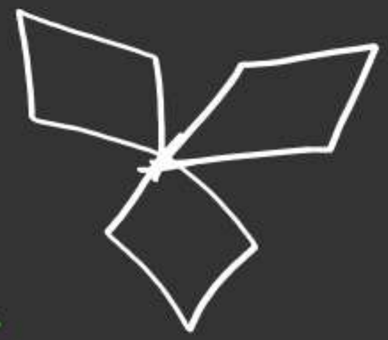
7

A	B	C
1	1	1

$$\begin{array}{r} 1 \\ -\ 1 \\ \hline 0\,0 \end{array}$$

$$\begin{array}{r} 0 \\ -\ 1 \\ \hline 1\,1 \end{array}$$

Step 3:- Logical expr



Difference $= \sum m(1,2,4,7) = A \oplus B \oplus C$

Borrow $= \sum m(1,2,3,7)$

$= 001 \quad 010 \quad 011 \quad 111$

$\bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC$

$= (\bar{A}\bar{B} + AB)C + \bar{A}B(\bar{C}+C)$

$= (\overline{A \oplus B})C + \bar{A}B \longrightarrow$ semi-minimized

$\bar{A}B + BC + \bar{A}C$

Step 4:- minimize.

Diff $= A \oplus B \oplus C$

Borrow $= \sum m(1,2,3,7)$

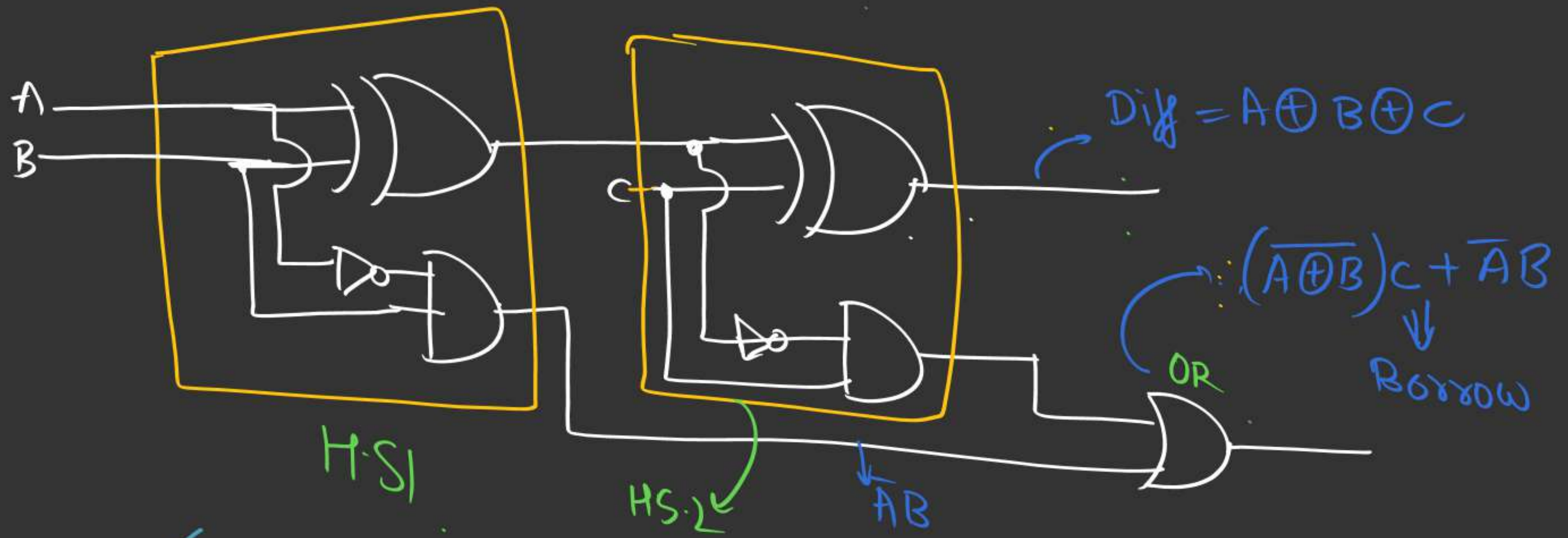
A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	(0)	1 (1)	1 (3)	1 (2)
A	(4)	(5)	1 (7)	(6)

$= (\bar{A}C + BC + \bar{A}B)$

semi-minimized : Imp obs :)

Step 5 : Hardware

$$\text{Borrow} = (\overline{A \oplus B})C + \overline{A}B$$



(1 Full Subtractor = 2 Half Subtractor + 1 OR gate)

Summary

HA	HS
FA	FS

Topic : Full Subtractor

HA

Sum = $A \oplus B$

Carry = AB

$\cancel{A} \to \bar{A}$

HS

Diff. $A \oplus B$

Borrow: $\bar{A}B$

FA

Sum = $A \oplus B \oplus C$

Carry = $AB + AC + BC$

FS

~~HS~~

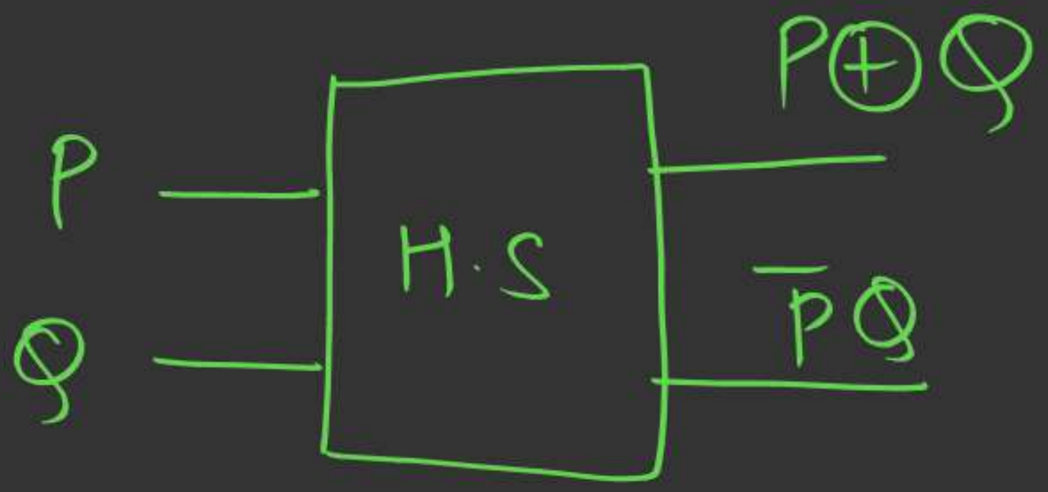
$A \to \bar{A}$

Diff. $A \oplus B \oplus C$

Borrow: $\bar{A}B + \bar{A}C + BC$

Imp Concept :

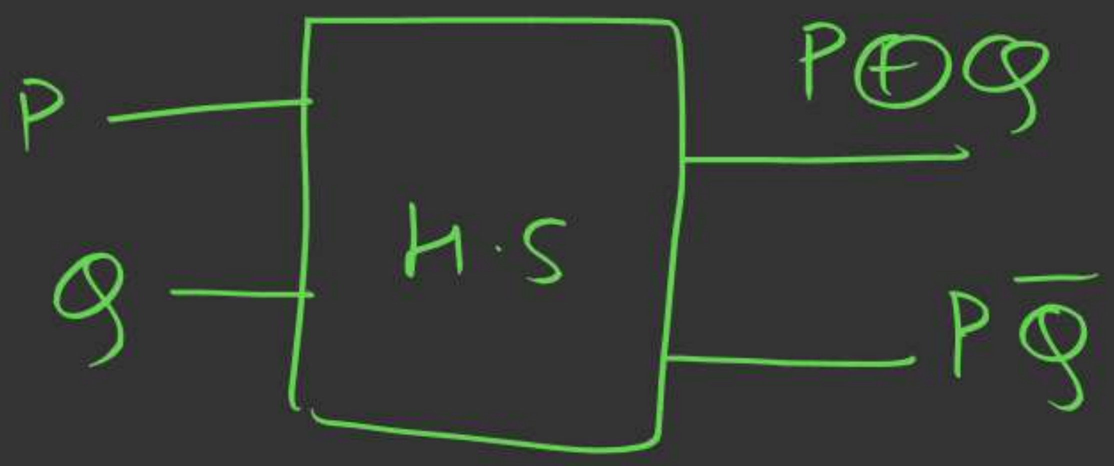
1



$\Downarrow$

$(P - Q)$

2

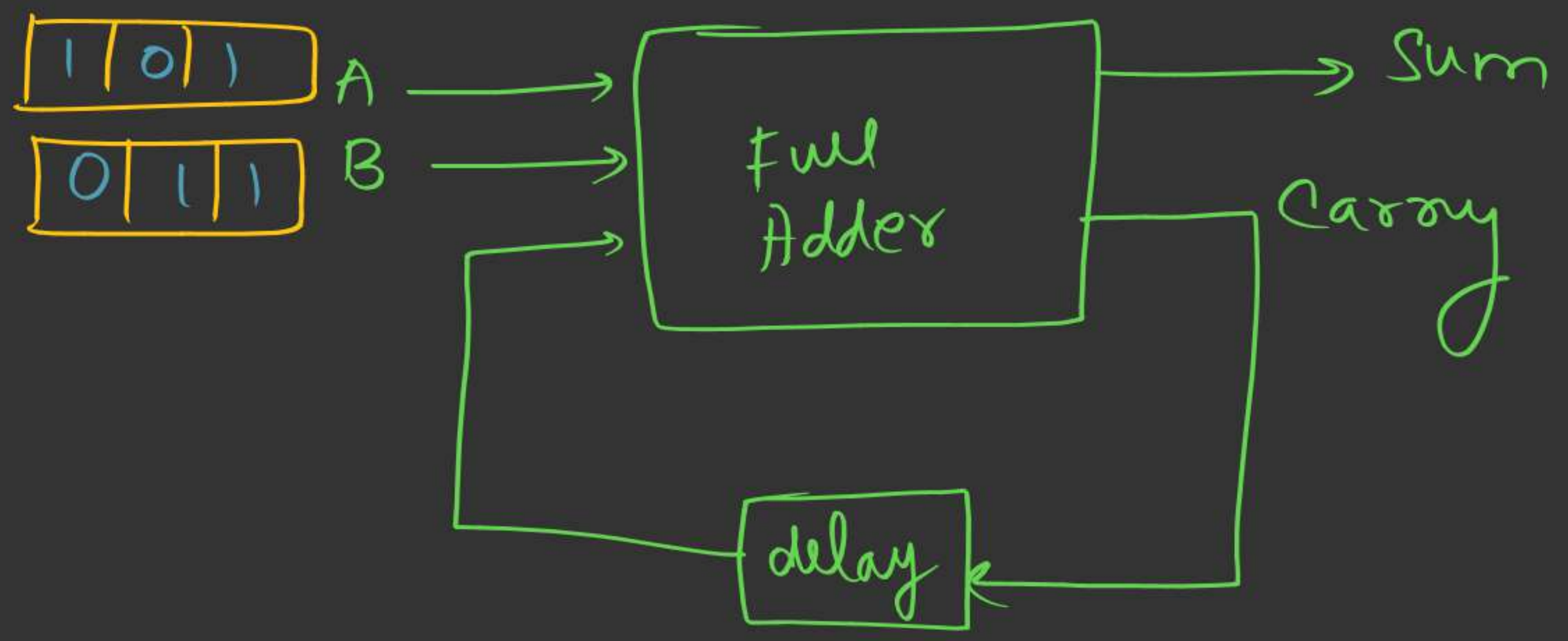


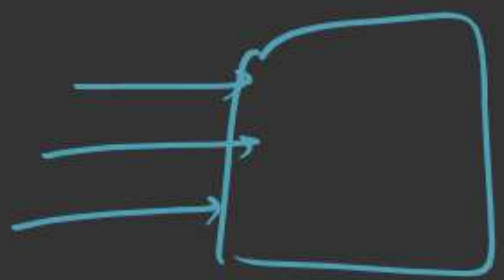
$\Downarrow$

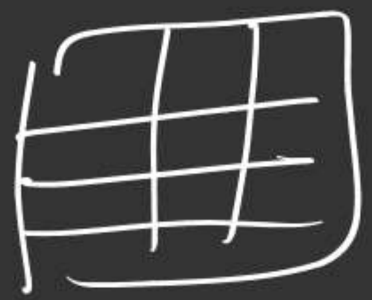
$(Q - P)$

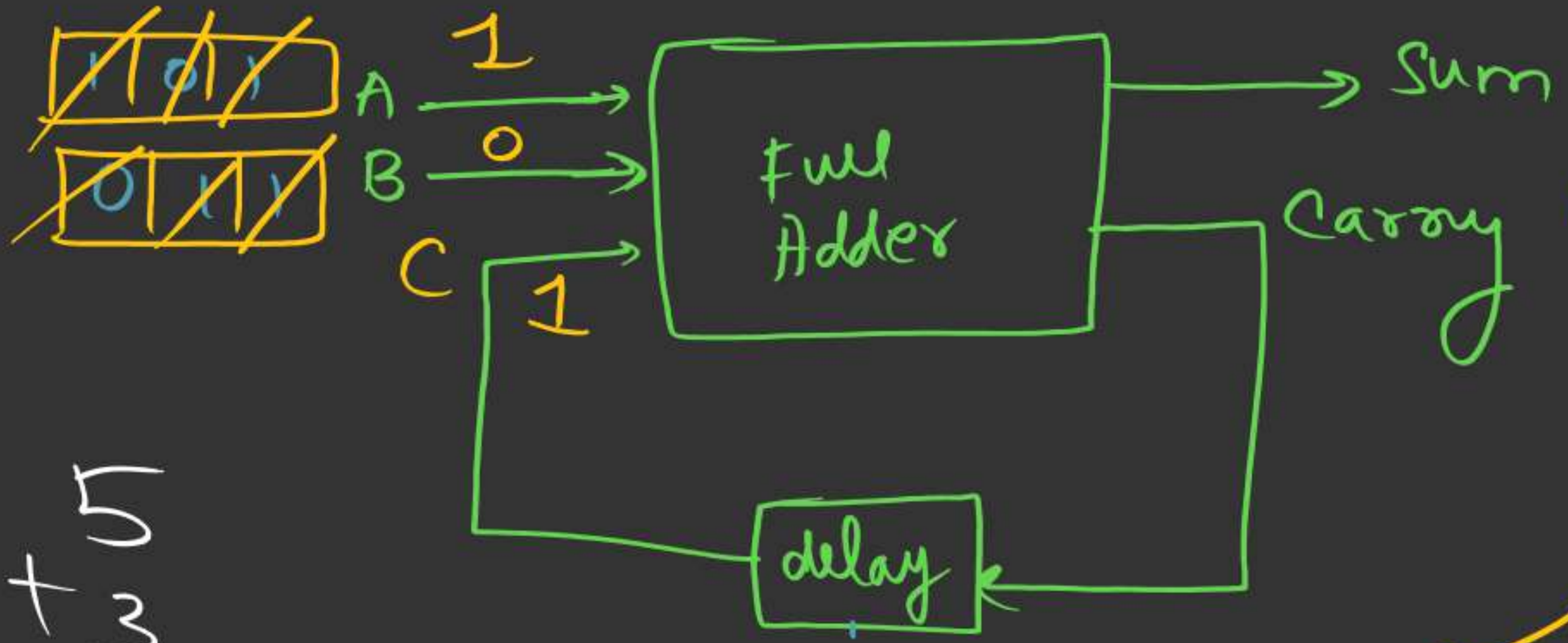
Serial Adder

Sequence / "one by one"


1 0 1
0 1 1
A
B
Full Adder
Sum
Carry
delay

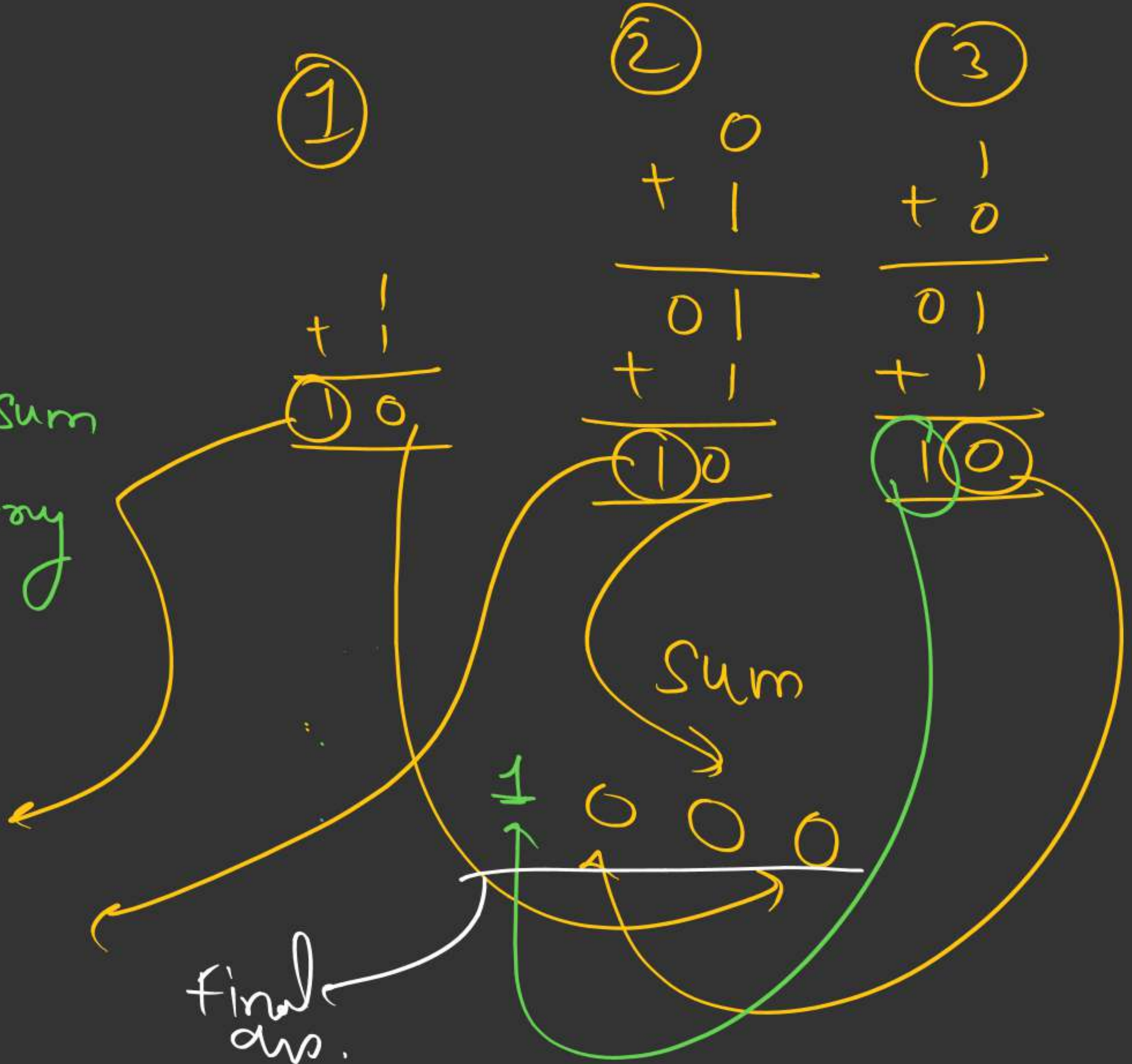






$$\begin{array}{r} 5 \\ +\ 3 \\ \hline 8 \end{array}$$

010



Working idea of Serial adder

MSB ← LSB

$$\begin{array}{ccc} 1 & 1 & 0 \\ 2 & 5 & 9 \\ 3 & 4 & 6 \\ \hline 6 & 0 & 5 \end{array}$$

1 1 0 → carry

6 0 5 → sum

Note:-

Delay is used for Synchronization purpose.

If T_d represents the delay of the Full adder.

then For a Serial Adder to add two n bits I/P, the time reqd is

$$\boxed{T = n * T_d}$$

Note:- Serial adder is the slowest adder among all the adders

Advanced Practice Questions

[MCQ] V. Adv

#Q. A logic circuit consist of two 2×4 decoders as shown in the figure.

The output of decoder are as follow:

$D_0 = 1$ when $A_0 = 0, A_1 = 0$

$D_1 = 1$ when $A_0 = 1, A_1 = 0$

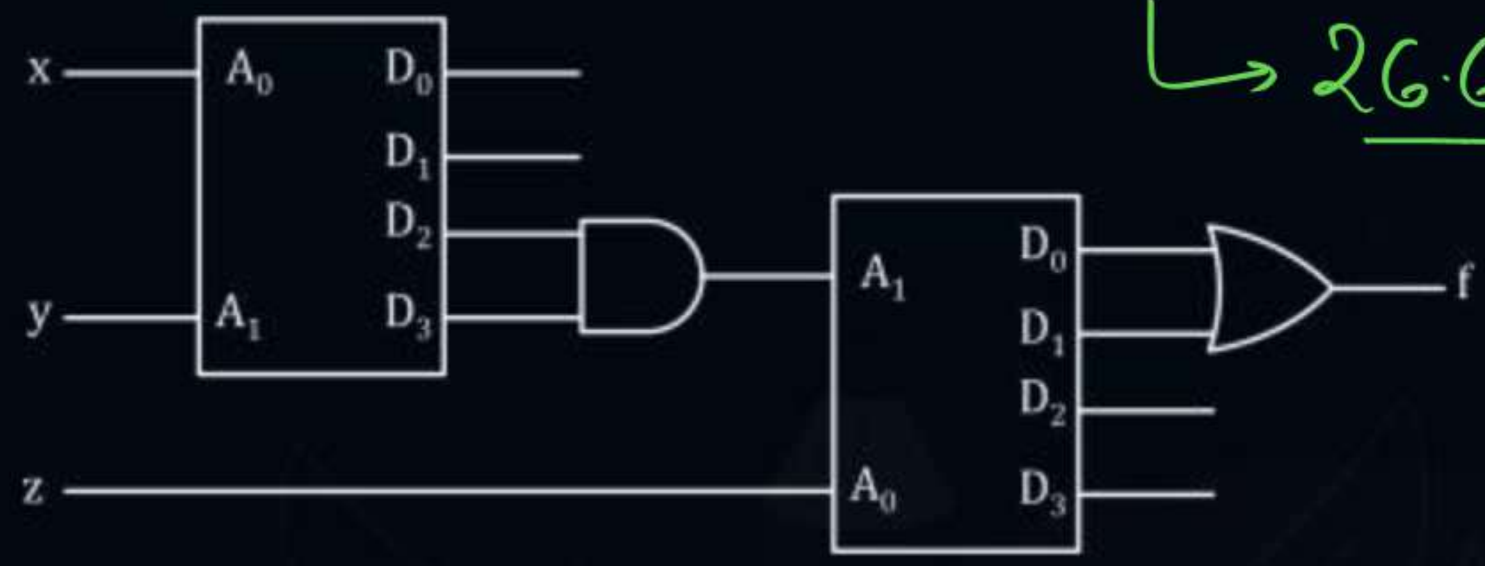
$D_2 = 1$ when $A_0 = 0, A_1 = 1$

$D_3 = 1$ when $A_0 = 1, A_1 = 1$

The value of f(x, y, z) is-

80%

↳ 26.6%



A) 0

B) $\bar{z}$

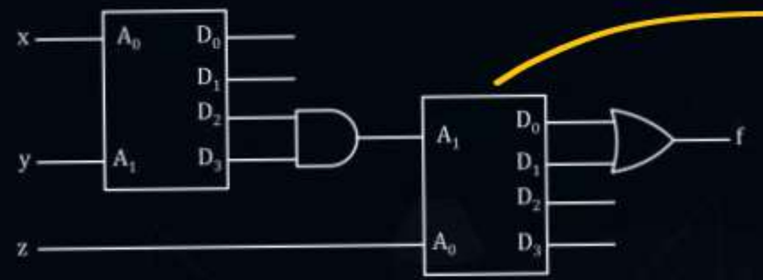
C) z

D) 1

[MCQ]

#Q. A logic circuit consist of two 2 × 4 decoders as shown in the figure.
The output of decoder are as follow:
$D_0 = 1$ when $A_0 = 0, A_1 = 0$
$D_1 = 1$ when $A_0 = 1, A_1 = 0$
$D_2 = 1$ when $A_0 = 0, A_1 = 1$
$D_3 = 1$ when $A_0 = 1, A_1 = 1$
The value of f(x, y, z) is-

2x4



A. 0

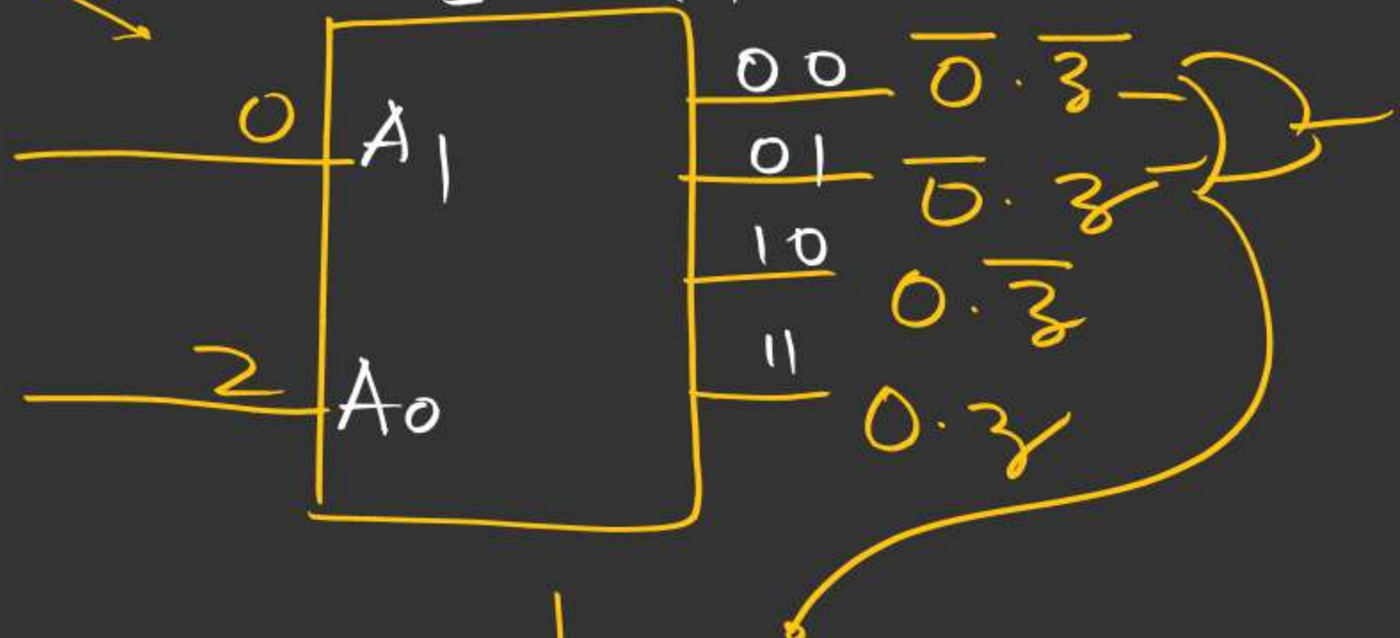
B. $\bar{z}$

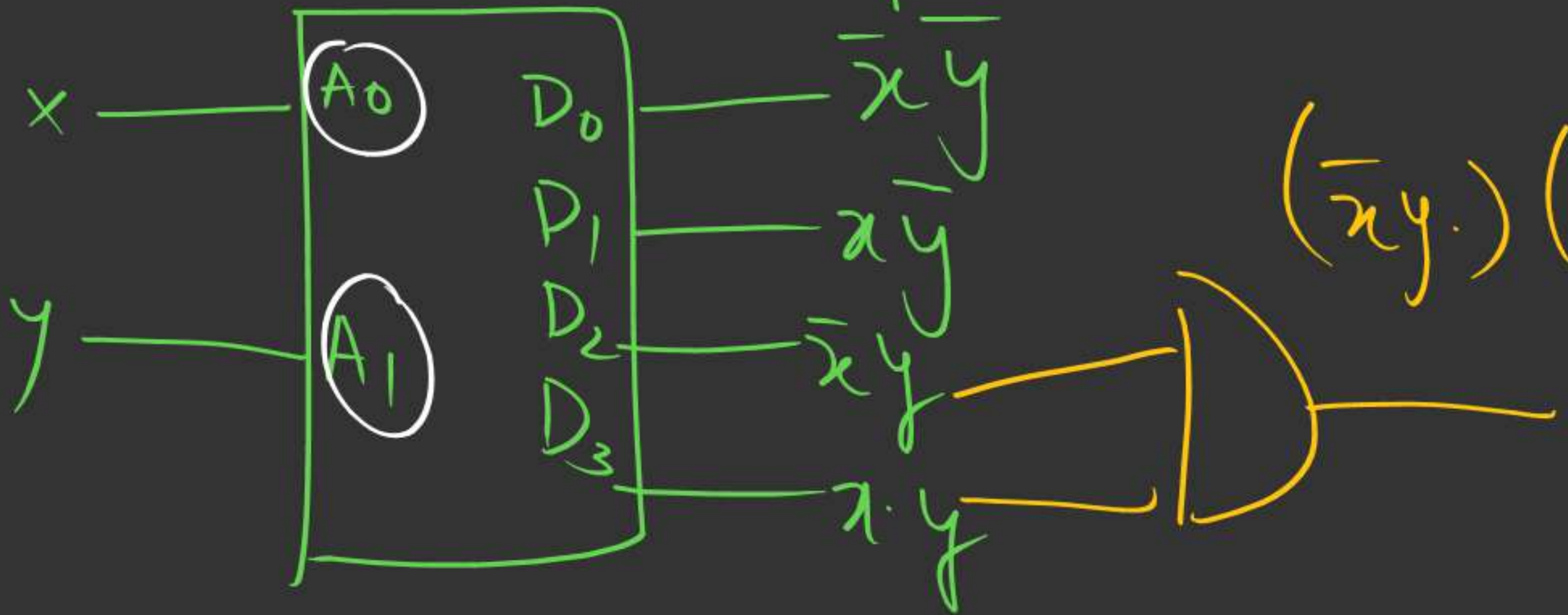
C. z

D. 1

(follow given)

$D_0 \rightarrow 0\ 0$ ($A_0\ A_1$)
$D_1 \rightarrow 1\ 0$
$D_2 \rightarrow 0\ 1$
$D_3 \rightarrow 1\ 1$





$(\bar{x}y)(xy) = 0$

Ans: $\bar{0}\bar{z} + \bar{0}z$
$= \bar{z} + z$
$= 1$

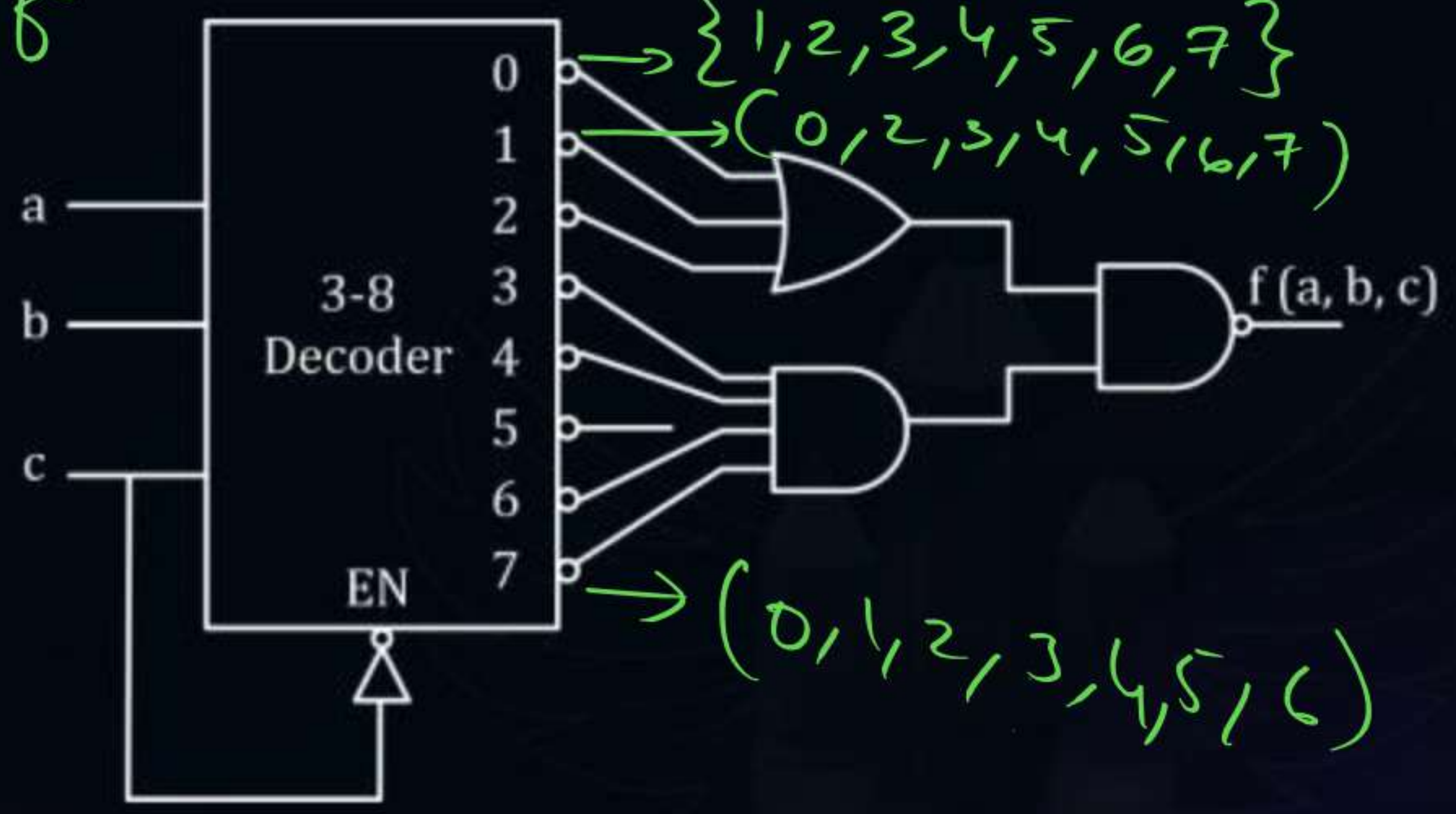
[MCQ]

V.adv

#Q. The Boolean expression f(a, b, c) in its canonical from for the decoder circuit shown below is:

follow this for ordering

33%



A $\Pi M\,(4, 6)$

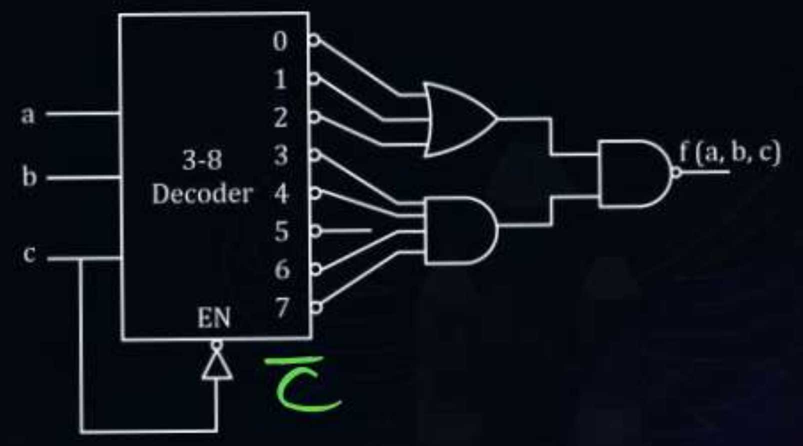
B $\sum m\,(0, 1, 2, 3, 5, 7)$

C $\sum m\,(4, 6)$

D $\Pi \sum\,(0, 1, 2, 3, 5)$

[MCQ]

#Q. The Boolean expression f(a, b, c) in its canonical from for the decoder circuit shown below is:



A. $\Pi M(4, 6)$

B. $\Sigma m(0, 1, 2, 3, 5, 7)$

C. $\Sigma m(4, 6)$

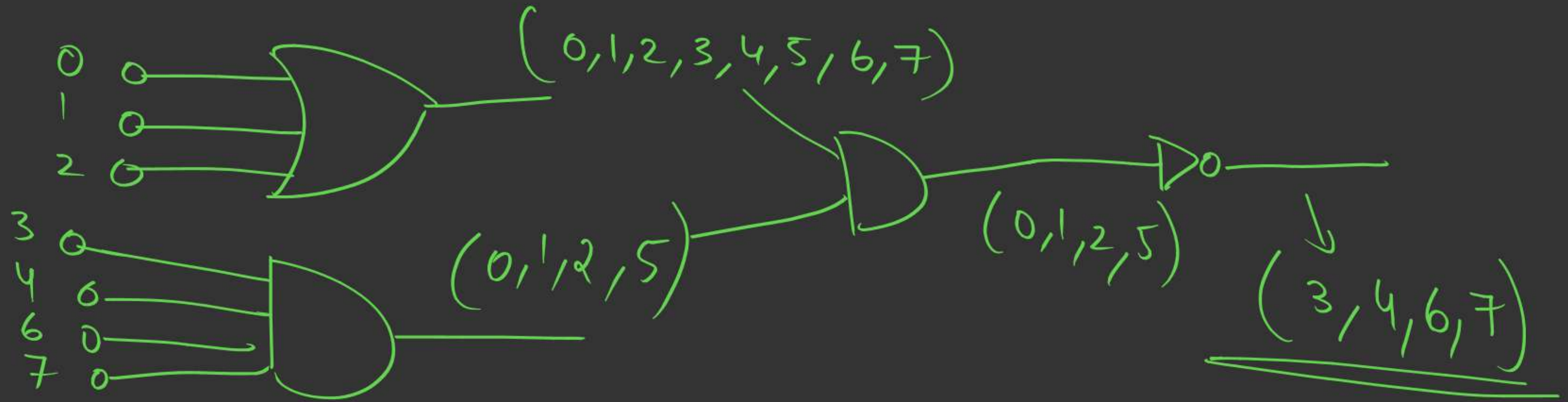
D. $\Pi\Sigma(0, 1, 2, 3, 5)$

Set

↓

OR → Union

AND → Intersection



Final answer

Enabler $= \overline{C}$

Final $m_o = \overline{C} \cdot \sum m(3,4,6,7)$

$\Downarrow$

$\sum m(4,6)$

3

4

6

7

[MCQ]

#Q. What is the Boolean expression for the output f of the combinational logical circuit of NOR gates given below?

80%

↓ 66.6%

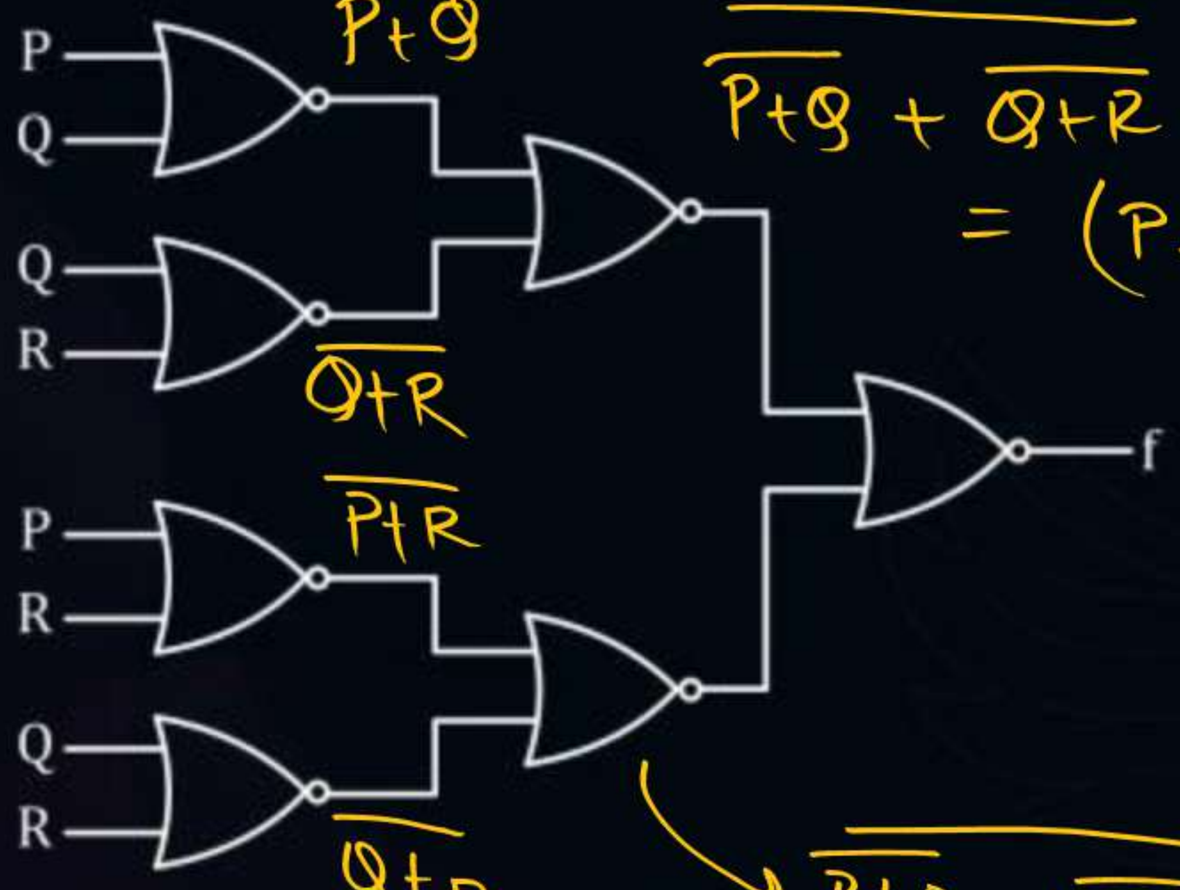
A $\overline{Q+R}$ ✓

B $\overline{P+Q}$ ✗

C $\overline{P+R}$ ✗

D $\overline{P+Q+R}$ ✗

0 1 0



$\overline{\overline{P+Q}+\overline{Q+R}}$

$= (P+Q)\cdot(Q+R)$

$\overline{\overline{P+R}+\overline{Q+R}} = (P+R)(Q+R)$

Simplify

$$\text{ans} = \overline{(P+Q)(Q+R) + (P+R)(Q+R)}$$

Let $\boxed{Q+R=X}$

$$= \overline{(Q+R)(P+Q+P+R)}$$

$$= \overline{(Q+R)(P+Q+R)}$$

$$= \overline{X(P+X)} = \overline{PX+X} = \overline{X(1+P)} = \overline{X}$$

$$= \overline{X} = \overline{Q+R}$$

2 mins Summary

HS ✓

FS ✓

Serial adder ✓

Adv Questions ✓

THANK - YOU